

MOLECULAR BASIS OF INHERITANCE

(Dept. of Zoology GIISS Mylachal)

DNA (Deoxyribo nucleic acid) and RNA (Ribonucleic acid) are the two types of nucleic acids present in the living organisms. DNA is the genetic material in most of the organisms. RNA is the genetic material in some viruses.

DNA

- DNA is the polymer of Deoxy ribo nucleotides.
- The length of DNA is expressed in number of nucleotides present.
- The number of nucleotides is characteristic of an organism.

Example

- | | | |
|-------------------------|-----|----------------------------------|
| 1. Bacteriophage 174 | has | 5386 nucleotides (base pairs-bp) |
| 2. Bacteriophage lambda | has | 48502 bp |
| 3. Escherichia coli | has | 4.6×10^6 bp |
| 4. Human haploid DNA | has | 3.3×10^9 bp |

Structure of a nucleic acid

- Nucleic acids are formed of nucleotides.
- A nucleotide has 3 parts – a nitrogenous base, a pentose sugar (ribose in RNA and deoxyribose in DNA) and a phosphate group.
- There are two types of nitrogenous base – purines and pyrimidines

Purine bases – adenine and guanine are purine bases.

Pyrimidine bases – cytosine, thymine and uracil are pyrimidine bases.

- Cytosine is present in both DNA and RNA
- Thymine is present only in DNA and Uracil is present only in RNA (at the place of thymine).
- Nitrogenous base is linked to the pentose sugar through N-glycosidic linkage to form nucleoside.

Eg. Adenine + ribose → Adenosine

Guanine + ribose → Guanosine

Cytosine + ribose → Cytidine

Thymine + ribose → Thymidine

Uracil + ribose → Uridine

Adenine + deoxy ribose → DeoxyAdenosine

Guanine + deoxy ribose → DeoxyGuanosine

Cytosine + deoxy ribose → DeoxyCytidine

Thymine + deoxy ribose → DeoxyThymidine

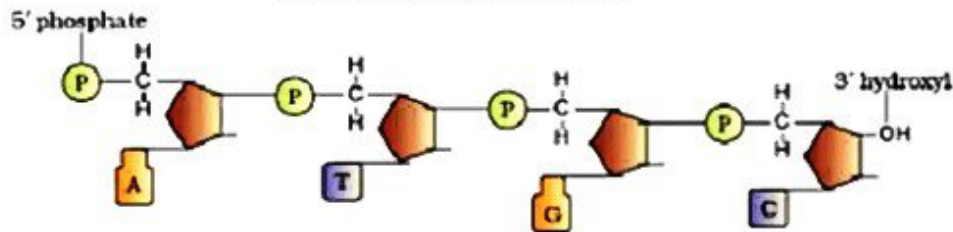
Uracil + deoxy ribose → DeoxyUridine

- A nucleotide is formed by the linkage of a nucleoside with a phosphate group through phosphoester linkage.

Nucleoside + phosphate → nucleotide

- Phosphate groups are linked to the 5th and 3rd carbon atom of ribose sugar to form a dinucleotide (5'-3' phosphodiester linkage)
- More nucleotides joined such manner to form a polynucleotide chain.
- A polymer thus formed has a free phosphate moiety at 5'-end of ribose sugar known as 5'-end of polynucleotide chain.
- The other end has a free hydroxyl group (OH) at 3'-end known as 3'-end of polynucleotide chain.
- The backbone of polynucleotide chain is sugar and phosphate.
- The nitrogenous bases linked to sugar moiety project from the backbone .

A polynucleotide chain



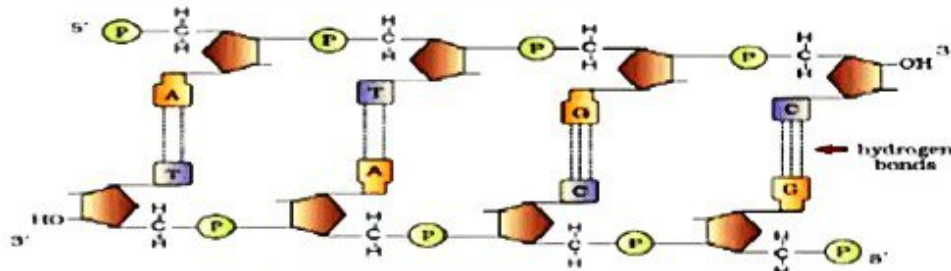
Double helical structure of DNA

- DNA was first identified by Freidrich Meisher in 1869 and he named it as Nuclein
- In 1953 James Watson and Francis Crick proposed double helix model for the structure of DNA.

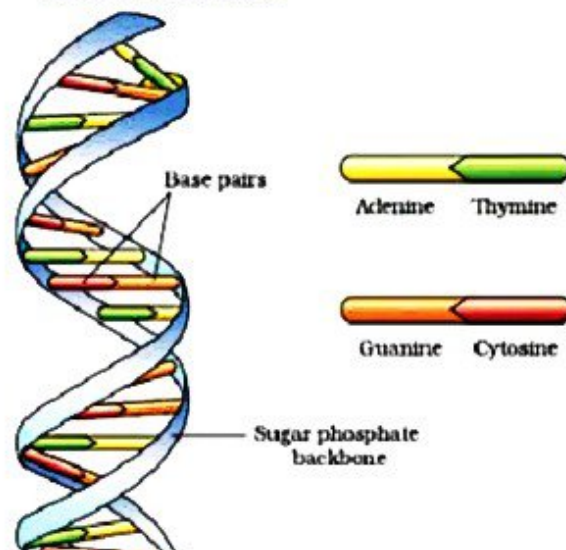
Salient features of the double helix structure of DNA

1. It is made of two polynucleotide chains. Where the backbone is sugar and phosphate, and the bases project inside.
2. The two chains have anti-parallel polarity. It means if one chain has the polarity $5' \rightarrow 3'$, the other has $3' \rightarrow 5'$
3. The bases in two strands are paired through hydrogen bond forming base pairs. Adenine forms two hydrogen bonds with Thymine. Guanine is bonded with Cytosine with three hydrogen bonds.
4. The two chains are coiled in a right handed fashion. The pitch of the helix is 3.4 nm and there is roughly 10 bp in each turn. So the distance between a bp in a helix is equal to 0.34 nm.

Double stranded polynucleotide chain



DNA double helix



Central dogma of molecular biology

- The DNA contains the genetic information for the cellular functions, development and heredity of organisms.
- Genes act by producing proteins.
- The unidirectional flow of genetic information from nucleic acid to protein (DNA to RNA and RNA to protein) is called central dogma of molecular biology. This was proposed by **Francis Crick**.

In some viruses (eg. HIV) the flow of information is in reverse direction i.e., from RNA to DNA. This is known as **reverse transcription**. (RNA → DNA → Protein)

Packaging of DNA double helix

- Taken the distance between two adjacent base pairs as 0.34 nm ($0.34 \times 10^{-9} \text{m}$), if the length of DNA double helix in a typical mammalian cell is calculated (simply by multiplying the total number of bp with distance between two adjacent bp, that is $6.6 \times 10^9 \text{ bp} \times 0.34 \times 10^{-9} \text{m/bp}$).
- It comes out to be approximately 2.2 meters. A length that is far greater than the dimension of a typical nucleus (approximately 10^{-6}m).

Arrangement of DNA in a prokaryotic chromosomes

- Prokaryotic cells lack a nuclear membrane and defined nucleus, even though DNA is not scattered throughout the cell.
- DNA, being negatively charged is held with some proteins having positive charges in a region termed as **nucleoid**.
- The DNA in nucleoid is organized in large loops held by proteins.

Arrangement of DNA in a eukaryotic chromosomes

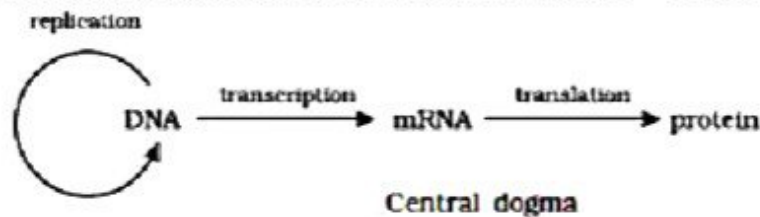
- In eukaryotes, the DNA is arranged around a set of positively charged basic proteins called **histones**.
- Histones are organized to form a unit of eight molecules called, **histone octamer**.
- The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called **nucleosome**.

Nucleosome

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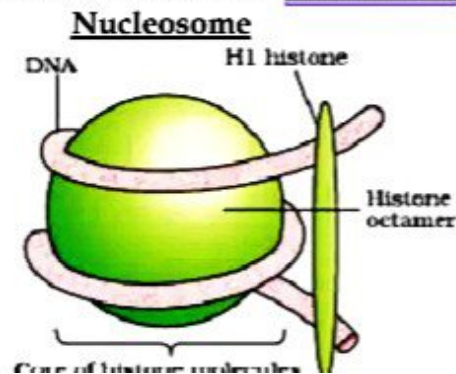
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- A typical nucleosome contains 200 -bp of DNA helix.
- Nucleosomes constitute the repeating unit of structure in nucleus called chromatin.
- The nucleosomes in chromatin are seen as 'beads on strings' structure under electron microscope.

NHC Proteins

- The packaging of chromatin at higher level requires additional set of proteins called Non histone chromosomal proteins (NHC proteins)

In a typical nucleus there are two types of chromatin,

Euchromatin and heterochromatin

Euchromatin

In a typical nucleus, some regions of chromatin are loosely packed and lightly stained known as euchromatin. This region contains active chromatin.

Heterochromatin

The chromatin that is more densely packed and stains dark are called heterochromatin. This region contains inactive chromatin.

TRANSFORMING PRINCIPLE (Griffith's Transformation Principle)

- The first scientist to observe transformation in bacteria was Frederick Griffith in 1928.
- He carried out experiments on a bacterium called *Streptococcus pneumoniae* (bacterium causes pneumonia).

Experiment

When *Streptococcus pneumoniae* (pneumococcus) bacteria are grown on a culture plate, some produce smooth shiny colonies (S) while others produce rough colonies (R). This is because the S strain bacteria have a mucous (polysaccharide) coat, while R strain does not. Mice infected with the S strain (virulent) die from pneumonia infection but mice infected with the R strain do not develop pneumonia.

S strain → Inject into mice → Mice die

R strain → Inject into mice → Mice live

Griffith was able to kill bacteria by heating them. He observed that heat-killed S strain bacteria injected into mice did not kill them. When he

S strain (heat-killed) → Inject into mice → Mice live

S strain (heat-killed)
+
R strain (live) → Inject into mice → Mice die

injected a mixture of heat-killed S and live R bacteria, the mice died. Moreover, he recovered living S bacteria from the dead mice.

He concluded that the R strain bacteria had somehow been **transformed** by the heat-killed S strain bacteria. Some 'transforming principle', transferred from the heat-killed S strain, had enabled the R strain to synthesise a smooth polysaccharide coat and become virulent. This must be due to the transfer of the genetic material. However, the biochemical nature of genetic material was not defined from his experiments.

Biochemical characterization of transforming principle

- Oswald Avery, MacLeod, Maclyn McCarty conducted many experiments to know the biochemical nature of transforming principle in Griffith's experiment.

They purified biochemicals (proteins, DNA, RNA, etc.) from the heat-killed S cells to see which ones could transform live R cells into S cells. They discovered that DNA alone from S bacteria caused R bacteria to become transformed.

They also discovered that protein-digesting enzymes (proteases) and RNA-digesting enzymes (RNases) did not affect transformation, so the transforming substance was not a protein or RNA. Digestion with DNase did inhibit transformation, suggesting that the DNA caused the transformation. They concluded that DNA is the hereditary material, but not all biologists were convinced.

'Hershy and chase experiment' to prove DNA as the genetic material

- Alfred Hershy and Martha Chase in 1952 proved that DNA is the genetic material.
- They worked with viruses that infect bacteria called **bacteriophages**.
- When a virus attaches with a bacteria, the genetic material of virus enters into the bacterial cell.
- The bacterial cell treats the viral genetic material as its own and more virus particles are produced.
- Hershy and Chase worked to discover whether the protein coat or DNA of the virus enters the bacterium.

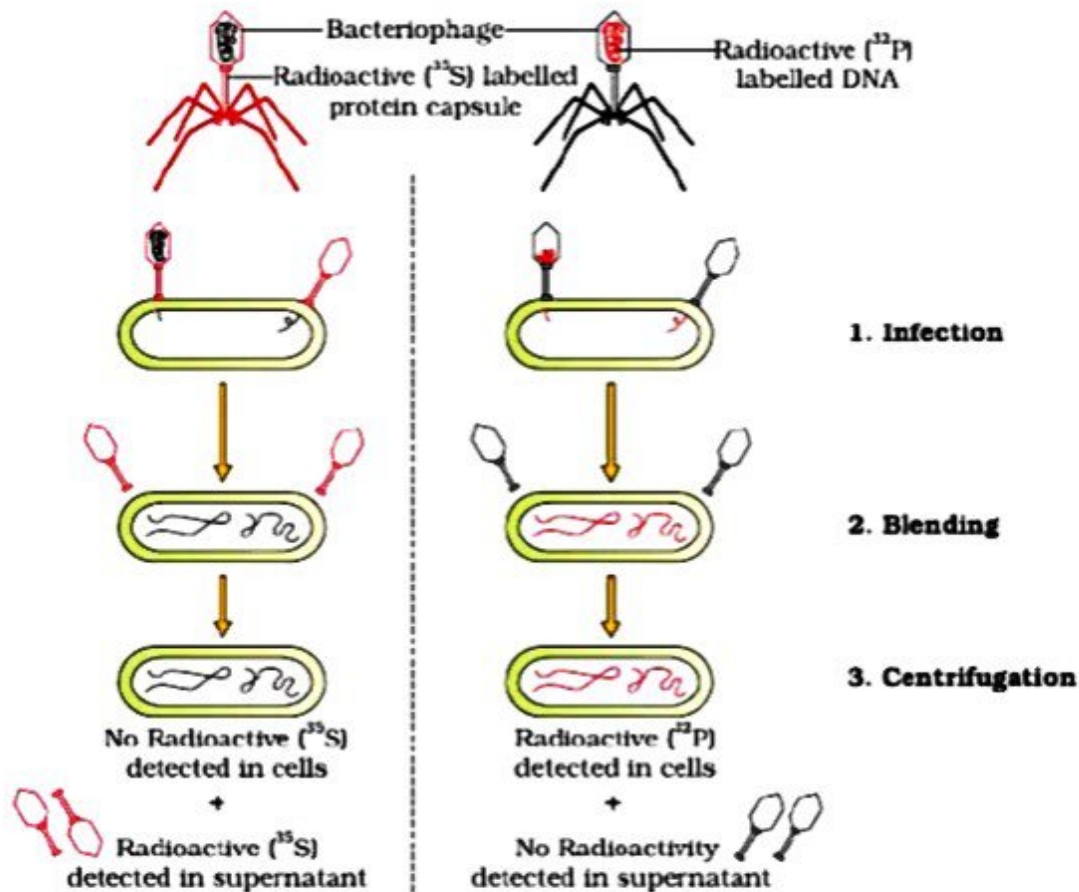
Experiment

- They grew some viruses in a radioactive medium of phosphorus and some others in a radioactive sulfur medium.
- Viruses grown in radioactive phosphorus medium have radioactive DNA.
- Viruses grown in radioactive sulfur medium have radioactive protein.
- These radioactive phages were allowed to attach to E.coli bacteria.
- After infection, the viral protein coats were removed from the bacteria.
- Bacteria which were infected with viruses with radioactive protein were not radioactive.

- But the bacteria which were infected with radioactive DNA were radioactive indicating that DNA was the material that passed from the virus to the bacteria.

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The Hershy Chase Experiment



Properties of Genetic material (DNA versus RNA)

- Hershy Chase experiment established that DNA is the genetic material. But in some viruses like Tobacco Mosaic Viruses (TMV) RNA is the genetic material.

Even though both RNA and DNA functions as genetic material, DNA is more stable for storage of genetic information than RNA because of the following reasons.

- The two strands of DNA are complementary to each other and can be separated by heating and come together in suitable conditions.
- 2'-OH group present in RNA is a reactive group and makes RNA labile and easily degradable.
- The absence of the 2'-OH group gives more stability and chemically less reactive nature to DNA than RNA.
- The presence of thymine at the place of uracil gives additional stability to DNA.
- Both DNA and RNA are able to mutate. In fact, RNA being unstable mutates at a faster rate. (RNA viruses have shorter life span due to faster mutations)

RNA World

- RNA was the first genetic material.
- RNA is essential for various life processes. It acts as genetic material as well as catalysts in some important biochemical reactions in living systems. But RNA being

a catalyst was reactive and hence unstable. Therefore, DNA has evolved from RNA with chemical modifications that make it more stable.

REPLICATION

- The process by which the synthesis of new DNA molecule from pre-existing DNA is called replication.
- DNA replicates during the synthetic phase of the interphase of cell cycle.

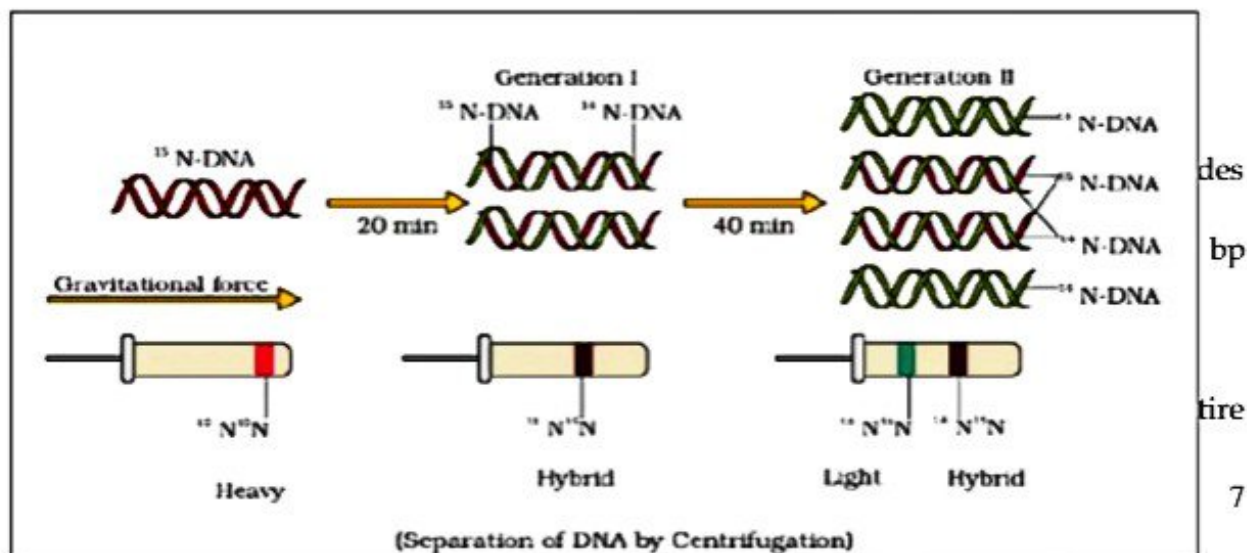
Watson and Crick proposed semiconservative mode of DNA replication

- In semiconservative replication, the newly formed DNA has one old strand and one new strand.
- The two strands separate and act as template for the synthesis of new complementary strands.
- After the completion of replication, each DNA molecule would have one parental and one newly synthesized strand.

Experimental proof for semiconservative replication

(Meselson and Stahl Experiment)

- Mathew Meselson and Franklin Stahl demonstrated the semiconservative mode of DNA replication in E.coli(1958).
- They cultured E.coli in a medium containing nitrogen salts labeled with the heavy isotope of nitrogen N^{15} . N^{15} was incorporated into both strands of DNA in the bacterium and thus became heavier. This DNA is called **heavier DNA**.
- Another preparation was also made, in this medium containing nitrogen salts labeled with N^{14} . N^{14} was also incorporated into both strands of DNA of E.coli.
- This DNA was lighter than the DNA of E.coli grown in N^{15} medium.
- Meselson and Stahl took E.coli cells from N^{15} medium and transferred to a N^{14} medium.
- After one generation, they isolated and centrifuged the DNA. Its density was intermediate between N^{15} DNA and N^{14} DNA.
- This shows that in the newly formed DNA one strand is **old(N^{15} type)** and one strand is **new(N^{14} type)**.
- DNA extracted from the culture after another generation was composed of equal amounts of this hybrid DNA and light DNA. This further confirmed semi-conservative replication.



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bp

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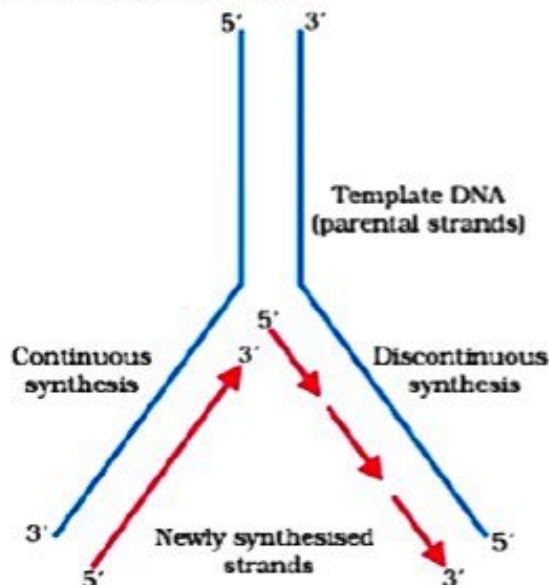
- In such DNAs replication occur within a small opening of the DNA helix known as replication fork.

Leading strand

- The DNA dependent DNA Polymerases catalyses polymerization in $5' \rightarrow 3'$ direction only.
- In $5' \rightarrow 3'$ direction, (the template with polarity $3' \rightarrow 5'$) the replication is continuous and it is known as leading strand.

Lagging strand

- In the other strand, template with polarity $5' \rightarrow 3'$, the replication is discontinuous and is called lagging strand.



Okazaki fragments

Discontinuously synthesized DNA fragments are called okazaki fragments.

DNA ligase

The discontinuously synthesized okazaki fragments are later joined by the enzyme called DNA ligase.

Origin of replication

The definite regions in which replication originates are known as origin of replication.

TRANSCRIPTION

- The process of copying genetic information from one strand of the DNA into RNA is called transcription.
- In transcription only a segment of DNA and only one of the strands is copied into RNA.

Transcription unit

A transcription unit in DNA has three regions.

1. A promoter
2. The structural gene
3. A terminator

- The two strands of DNA have opposite polarity and the DNA dependent RNA Polymerase also catalyses the polymerisation in only one direction ie, $5' \rightarrow 3'$ direction.

- The strands that has the polarity 3'→5' acts as a template and is known as **coding strand, which** does not code for anything.

Eg.

3'-ATGCATGCATGCATGCATGC-5' Template Strand

5'-TACGTACGTACGTACGTACGTACG-3' Coding Strand

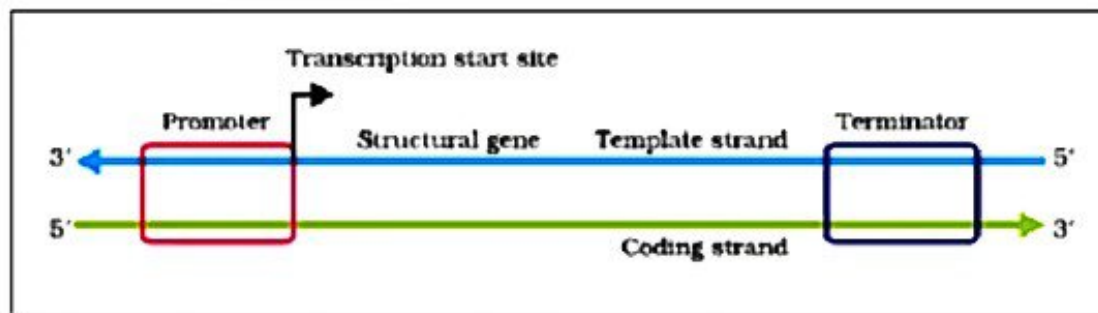
Promoter

- Promoter is a DNA sequence that provides binding site for polymerase. It is located towards the 5' end of the structural gene.
- The presence of a promoter in a transcription unit defines the template and coding strands.

Terminator

- Terminator is located towards 3' end of the coding strand and it defines the end of the process of transcription.

Schematic structure of a transcription unit



TRANSCRIPTION UNIT AND THE GENE

Cistron

Cistron is a segment of DNA coding for a polypeptide.

Monocistronic unit

- The structural unit in transcription unit of eukaryotes are monocistronic.
- Monocistronic structural genes have interrupted coding sequences. I.e, the genes are split or the genes with coding sequences and genes with non-coding sequences.

Polycistronic unit

This is the structural transcription unit of bacteria or prokaryotes.

Exons

The coding sequence or expressed sequences of cistron are called exons.

Introns

The non coding sequences of cistrons are called introns or intervening sequence. They do not appear in the mature or processed RNA.

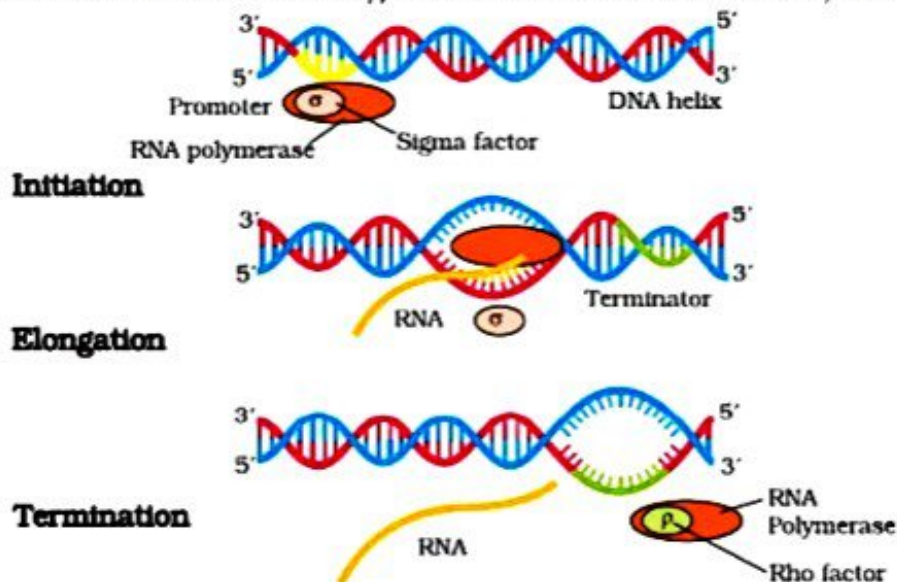
TYPES OF RNA AND THE PROCESS OF TRANSCRIPTION

- In bacteria there are three major types of RNA
 - Messenger RNA or m RNA** → acts as the template
 - Transfer RNA or t RNA** → brings amino acids and reads the genetic code
 - Ribosomal RNA or r RNA** → plays structural and catalytic role during translation
- All the 3 RNAs are needed for protein synthesis.

- A single DNA – depended RNA Polymerase catalyses transcription of all types of RNA in bacteria.

Process of transcription in bacteria(prokaryotes)

- The process of transcription includes 3 steps- **initiation** , **elongation** and **termination**.
- RNA Polymerase binds to promoter and initiates transcription.
- The binding of the RNA Polymerase causes local unwinding of the DNA double helix.
- Unwinding is followed by elongation.
- Elongation is also helped by the core enzyme , RNA Polymerase.
- When the polymerase reach the terminator region, the newly formed RNA and RNA Polymerase fall off. This results in termination of transcription.
- All the 3 steps were catalysed by RNA Polymerase. It is associated with **initiation factor (sigma factor)** and **termination factor (Rho factor)**
- In bacteria transcription and translation process takes place in the same compartment, because there is no separation of cytosol and nucleus in bacteria. Moreover the translation can begin much before the RNA is fully transcribed.



Process of transcription in eukaryotes has two additional complexities:-

1. In eukaryotes there are 3 RNA Polymerases-
 - a) RNA Polymerase 1 - Transcribes r RNAs (28s,18s,5.8s)
 - b) RNA Polymerase II - Transcribes precursor m RNAs and heterogeneous nuclear RNA (hnRNA)
 - c) RNA Polymerase III - Transcribes t RNA, 5s r RNA and snRNAs (small nuclear RNAs)
2. The primary transcript contains both the exons and the introns.
 - By the process of **splicing**, the introns are removed from hn RNA and exons are spliced(joined) together.
 - Then it is subjected to **capping and tailing**.

Capping

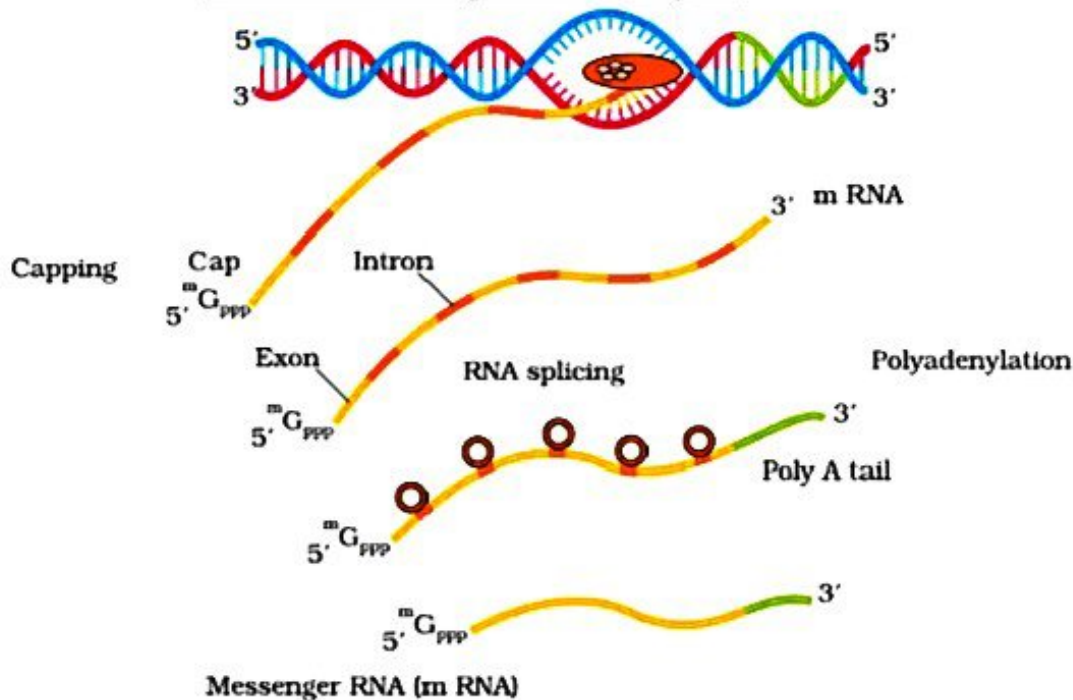
In capping an unusual nucleotide (**methyl guanosine triphosphate**) is added to the 5' end of hn RNA.

Tailing

In tailing, **adenylate residues (200-300)** are added at 3' end.

Now the hnRNA is fully processed and it is called m RNA , which is transported out of the nucleus for translation.

Process of transcription in eukaryotes



GENETIC CODE

- The sequence of nucleotides (nitrogen bases) in the mRNA which contains the information for protein synthesis is known as genetic code.
- The genetic information is written in a coded language in the form of **triplet codons (3 letter code)**
- It is first transcribed into mRNA and then translated into protein.
- The sequence of three bases determining a single amino acid is called **codon**
- It was **George Gamow**, a physicist who argued that since there are only 4 bases and if they have to code for 20 amino acids, the code should constitute a combination of 4 bases.
- In the light of this argument, he suggested that in order to code for 20 amino acids, the code should be made up of three nucleotides.
- This would generate 64 codons (4^3 i.e., $4 \times 4 \times 4 = 64$) for coding 20 amino acids.
- Of the 64 codons, 61 are sense codons and 3 are **non-sense codons**.
- The non-sense codons do not code for any amino acids. They are **UAA, UGA, and UAG**.

The genetic code dictionary

Ala	Alanine	Asp	Aspartic acid	Glu	Glutamic acid
Ile	Isoleucine	Met	Methionine	Ser	Serine
Tyr	Tyrosine	Arg	Arginine	Cys	Cystine
Gly	Glycine	Leu	Leucine	Phe	Phenyl alanine
Thr	Threonine	Val	Valine	Asn	Asparagine
Gln	Glutamine	His	Histidine	Lys	Lysine
Pro	Proline	Trp	Tryptophan		

The codons for various amino acids

First position	Second position				Third position
	U	C	A	G	
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

The salient features of genetic code

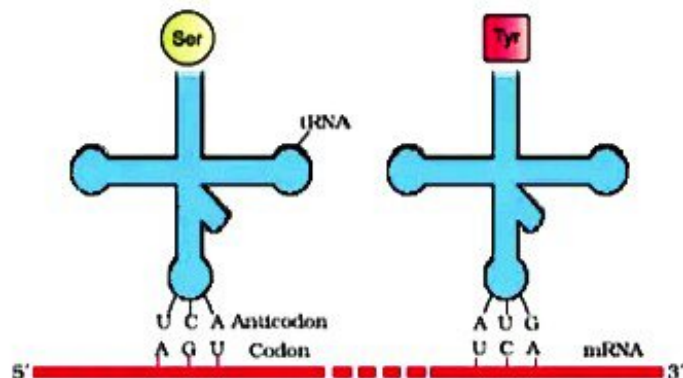
- (i) The codon is triplet. 61 codons code for amino acids and 3 codons do not code for any amino acids, hence they function as stop codons.
- (ii) One codon codes for only one amino acid, hence, it is **unambiguous** and **specific**.
- (iii) Some amino acids are coded by more than one codon, hence the code is **degenerate**.
- (iv) The codon is read in mRNA in a contiguous fashion. There are no punctuations.
- (v) The code is nearly **universal**: for example, from bacteria to human UUU would code for Phenylalanine (phe). Some exceptions to this rule have been found in mitochondrial codons, and in some protozoans.
- (vi) AUG has dual functions. It codes for Methionine (met) , and it also act as **initiator** codon.

MUTATIONS AND GENETIC CODE

- We can study the relationship between the genes and DNA by mutation studies .
- Mutations affect the functions of gene.
- The changes occurring the structure of a gene are called **point mutations or gene mutations.**
- Mutations alter the information in the gene as well as the message for protein synthesis.
- Insertion or deletion of one or two bases changes the reading frame. Such mutations are known as **frame shift mutations or deletion mutations.**

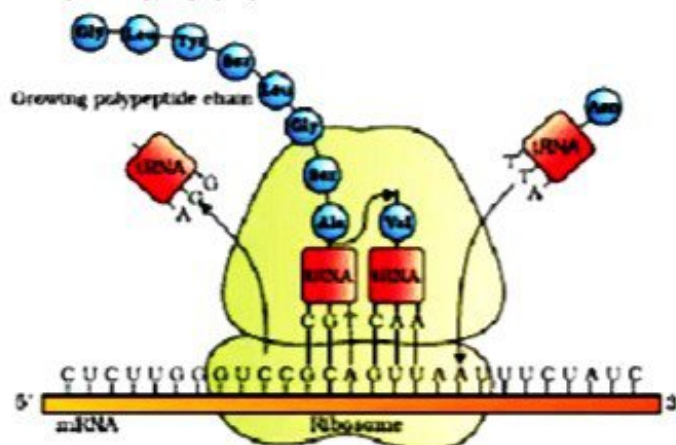
t-RNA - the Adapter Molecule

- The t RNA plays an important role in protein synthesis.
- They are the smallest RNA molecules and found in the cytoplasm of the cell.
- Before the genetic code was postulated, the t RNA was called as s RNA or soluble RNA.
- Each t RNA has a triplet code which is complementary to a codon in the m RNA called **anticodon**.
- It also has an amino acid acceptor end to which it binds to amino acids.
- For initiation, there is specific t RNA called initiator t RNA.
- The secondary structure of t RNA looks like clover leaf. In actual structure t RNA looks like inverted L.



TRANSLATION

- It is the process of polymerization of amino acids to form a polypeptide.
- The sequence of bases in the m RNA determines the order and sequence of amino acids in a protein.
- The amino acids in a polypeptide are linked by peptide bonds.
- So the first process of protein synthesis is the activation of amino acids with t RNA in the presence of ATP. This is called as charging of t RNA or aminoacylation of t RNA.
- Ribosome is the site of protein synthesis.
- In an inactive stage, it exists as two subunits- a **large subunit** and a **small subunit**.
- The process of translation begins when the small subunit encounters the m RNA.
- The large subunit has two binding site for t RNA (A site -aminoacyl t RNA binding site and P site-peptidyl site)
- The ribosome also acts as a catalyst for the formation of peptide bond.
- The initiation codon for methionine is AUG, the methionyl t RNA complex would have UAC at the anticodon site.
- The initiating t RNA is found at the P site. All other t RNAs first bind to the A site and then shift to the P site.
- An m RNA also has some additional sequences that are not translated and are referred as untranslated region(UTR). The UTRs are present at both 5' end(before start codon) and at 3' end (after stop codon). UTRs are essential for efficient translation process.
- For initiation, the ribosome binds to the m RNA at the start codon, AUG.
- The ribosome proceeds to the elongation phase of protein synthesis. During this stage, complexes composed of an amino acid linked to t RNA, sequentially bind to the appropriate codon in m RNA by forming complementary base pairs with the t RNA anticodon.
- The ribosome moves from codon to codon allowing the m RNA. Amino acids are added one by one, translated into polypeptide sequences dictated by DNA and represented by m RNA.
- At the end, a release factor binds to the stop codon, terminating translation and releasing the complete poly peptide from the ribosome.



REGULATION OF GENE EXPRESSION

- The gene expression results in the formation of a polypeptide
- It can be regulated at the following levels in eukaryotes
 - (i) **transcriptional level (formation of primary transcript).**
 - (ii) **processing level (regulation of splicing).**
 - (iii) **transport of mRNA from nucleus to the cytoplasm,**
 - (iv) **translational level.**

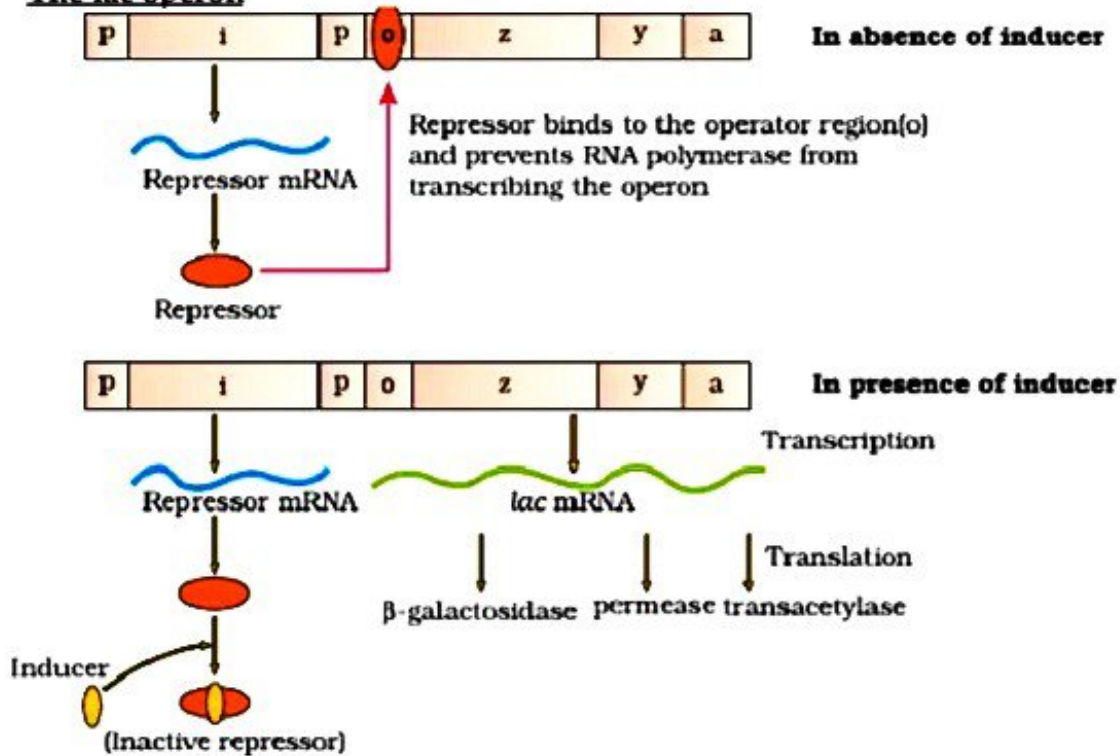
Example

- The enzyme **beta-galactosidase** is synthesized by E.coli to catalyse the hydrolysis of lactose into galactose and glucose.
- If the bacteria are living in medium devoid of lactose, they need not require the particular enzyme. That is, the gene expression is controlled by metabolic, physiological or environmental conditions.
- Similarly the development and differentiation of embryo into adult organisms are also a result of the co-ordinated activities of several set of genes and their expressions.

THE LAC OPERON

- This concept was proposed by **Francis Jacob and Jaques Monod** in 1961.
- This operon concept states that **each metabolic reaction is controlled by a set of genes.**
- All the genes regulating a metabolic reaction constitute an **operon**.
- The lac operon consists of one **regulatory gene (the 'I' gene) and 3 structural genes (z,y and a).**
- **The 'I' gene codes for the repressor of the lac operon.**
- **The 'z' gene codes for the beta galactosidase,** which hydrolyse lactose to glucose and galactose.
- **The 'y' gene codes for permease,** which increases permeability of the cell to beta galactosides.
- **The 'a' gene code for transacetylase.**
- Lactose is the substrate for the enzyme beta galactosidase and it regulates switching on and off the operon. Hence it is termed as **inducer.**
- In the absence of glucose, if lactose is present in the growth medium, bacteria transport the lactose with the help of permease.
- The repressor protein synthesized by the 'I' gene in the absence of inducer binds to the operator region of the operon and prevents RNA Polymerase from transcribing the operon.
- In the presence of inducer (lactose) the repressor is inactivated and RNA Polymerase begins to transcribe the gene 'z', 'y' and 'a'.
- Regulation of lac operon by repressor is referred to as **negative regulation**

The lac operon



HUMAN GENOME PROJECT (HGP)

- It is learnt that the genetic make up of an organism or an individual lies in the DNA sequences
- The DNA sequences are different at least at some places in different individuals.
- This led to the launching of human genome project to find out the complete DNA sequence of human genome.
- The entire DNA in the cells of an organism is known as genome.
- Human genome project was started in 1990.
- Human genome is said to have approximately 3×10^9 bp and if the cost of sequencing required is US \$ 3 per bp (the estimated cost in the beginning), the total estimated cost of the project will be around 9 billion US dollars. HGP was closely associated with the rapid development of a new area in biology called **bioinformatics**

Goals of Human Genome Project (HGP)

- Identify all the approximately 20,000-25,000 genes in human DNA;
- Determine the sequences of the 3 billion chemical base pairs that make up human DNA;
- Store this information in databases;
- Improve tools for data analysis;
- Transfer related technologies to other sectors, such as industries;
- Address the ethical, legal, and social issues (ELSI) that may arise from the project.

Methodologies

- There are two methods involved in HGP.
- The first method is focused on identifying all the genes that are expressed as RNA referred to as Expressed Sequence Tags (ESTs)
- The second is sequencing the whole set of genome that contained all the coding and non-coding sequence, the term which is referred to as Sequence Annotation.
- For sequencing, the total DNA from a cell is isolated and converted into fragments with smaller sizes and cloned in suitable host using specialized vectors.
- The purpose of cloning is amplification of each piece of DNA fragment that would make sequencing more easily.
- The commonly used hosts were called as BAC (Bacterial Artificial Chromosomes) and YAC(Yeast artificial chromosomes).
- The fragments were sequenced using automated DNA sequencers that worked on the principle of a method developed by Frederic K. Sanger.
- These sequences were arranged based on some overlapping regions present in them
- For alignment of these sequences, special computer based programmes were developed.
- With the help of this, the sequences were subsequently annotated.
- The sequence of chromosomes 1 was completed only in May 2006(this was the last of the 24 human chromosomes-22 autosomes and X and Y- to be sequenced).

Salient Features Of Human Genome

Some of the salient observations drawn from human genome project are as follows:

- (i) The human genome contains 3164.7 million nucleotide bases.
- (ii) The average gene consists of 3000 bases, but sizes vary greatly, with the largest known human gene being dystrophin at 2.4 million bases.
- (iii) The total number of genes is estimated at 30,000—much lower than previous estimates of 80,000 to 1,40,000 genes. Almost all (99.9 per cent) nucleotide bases are exactly the same in all people.
- (iv) The functions are unknown for over 50 per cent of discovered genes.
- (v) Less than 2 per cent of the genome codes for proteins.
- (vi) Repeated sequences make up very large portion of the human genome.
- (vii) Repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundred to thousand times. They are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution.
- (viii) Chromosome 1 has most genes (2968), and the Y has the fewest (231).
- (ix) Scientists have identified about 1.4 million locations where single-base DNA differences (**SNPs – single nucleotide polymorphism**, pronounced as 'snips') occur in humans. This information promises to revolutionise the processes of finding chromosomal locations for disease-associated sequences and tracing human history.

DNA FINGERPRINTING

- The technique which is used to identify the similarities of the DNA fragments of two individuals is called DNA Fingerprinting.
- It was first developed by Alec Jeffreys in 1985.
- In DNA Fingerprinting, the DNA molecules are identified by a technique known as southern blotting, developed by EM Southern in 1975.
- To take DNA Fingerprint, repetitive DNA is isolated. A major portion of this comprises a category of non-coding sequences called repetitive sequences.
- The repetitive sequences vary from person to person .
- These repeating sequences are also called Variable Number Tandem Repeats(VNTRs)
- Repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundred to thousand times.
- When the repetitive DNA sequences are separated from the bulk genomic DNA as different peaks during density gradient centrifugation, the bulk DNA forms major peak and the other small peaks referred to as satellite DNA)
- The human genome contains 3×10^9 (3 billion) nucleotides.

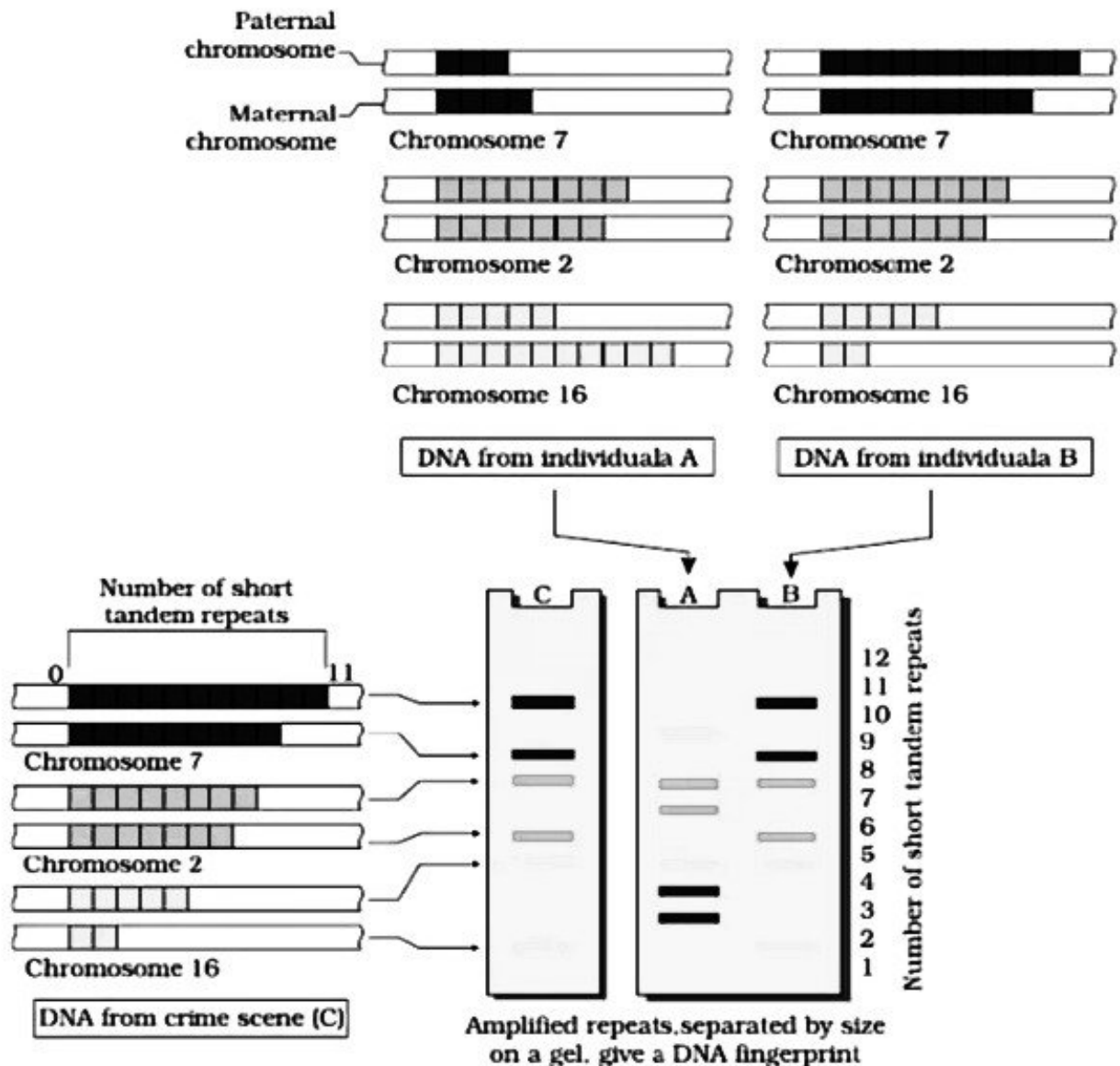
Steps involved in DNA Fingerprinting (Southern Blotting)

1. Isolation of DNA from body (blood, hair, semen etc.)
2. Then it is treated with restriction endonuclease enzyme to cut the repetitive DNA into fragments.
3. Gel Electrophoresis is used to separate the fragments.
4. The separated double stranded DNA fragments are then treated with alkali, NaOH. As a result the double stranded DNA (ds DNA) fragments become single stranded DNA(ssDNA) and denatured.
5. The single stranded DNA fragments are blotted on to a nitrocellulose filter paper and then baked in a vacuum oven at 80°C for 3-5 hours. This is to fix the DNA fragments on the membrane. This process is known as blotting.
6. The nitrocellulose filter paper is placed in a solution containing DNA Probe(the radioactive labeled single stranded DNA is called DNA probe). The DNA probe binds with the complementary sequences of the DNA fragments on the membrane to form a hybridized DNA. Then the filter paper is washed to remove the unbound probe.
7. The hybridized DNA is then photographed on to an X-ray film by autoradiography. The image of the hybrid DNA obtained by autoradiography is called DNA Fingerprint.

Applications of DNA Fingerprinting

- It is used as a powerful forensic tool to solve the problems of paternity, rape, murder etc.
- It is used in the diagnosis of genetic diseases.
- It is used in the determination of phylogenetic status of animals etc.

Schematic representation of DNA fingerprinting



Schematic representation of DNA fingerprinting: Few representative chromosomes have been shown to contain different copy number of VNTR. For the sake of understanding different colour schemes have been used to trace the origin of each band in the gel. The two alleles (paternal and maternal) of a chromosome also contain different copy numbers of VNTR. It is clear that the banding pattern of DNA from crime scene matches with individual B, and not with A.